Week : 8

الأسبوع #: من - إلى

Subject/Chemistry :الموضوع **Unit TitleReactivity 2.2 : عنوان الوحدة** **Grade: 12 (DP2)**

Lessonsالحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formativeتكويني/ Summativeختامي / Differentiationالتمايز	Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور
Lesson 1: الحصة 1 R2.2.1 Rate of reaction	**Learning target (DOK 2):** I can explain how the rate of reaction is expressed as the change in concentration of a reactant or product per unit time, and determine and compare rates from tangents to concentration, volume or mass against time graphs.	**Formative:** warm-up on mini whiteboards, hinge question, graph and tangent practice, and the exit ticket. **Differentiation:** extension questions on comparing rates and the NOS/TOK challenge for early finishers. **UDL strategies:** 3.1 Activate prior knowledge using organizers, links, and pre-teaching. 2.2 Clarify syntax/structure and highlight relationships in text, formulas, diagrams. 8.2 Vary challenge level with differentiated tools, resources, and difficulty.	Video: The rate of a chemical reaction, including graphs (Oak National Academy)
Lesson 2 الحصة الثانية R2.2.2 Collision theory	**Learning target (DOK 2):** I can explain how particle kinetic energy, temperature in kelvin and collision geometry determine whether a collision leads to a reaction.	**Formative:** retrieval starter, 'spot the mistake' on collision diagrams and a Must/Should/Could exit ticket. **Differentiation:** choose-your-challenge practice with self-marking. **UDL strategies:** 2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 3.3 Guide processing with explicit steps, chunking, and interactive models. 8.2 Vary challenge level with differentiated tools, resources, and difficulty.	Video: Collision theory and the Maxwell-Boltzmann distribution (Khan Academy)

Lesson 3 **الحصة الثالثة** **R2.2.3 Factors affecting rate**	**Learning target (DOK 2):** I can predict and explain, using collision theory, how pressure, concentration, surface area, temperature and a catalyst change the rate of a reaction.	**Formative:** mini-whiteboard predictions for each factor and an exit ticket. **Differentiation:** scaffolded prediction table for Must; explanations using collision theory for Should; controlled variables and graph error questions for Could. **UDL strategies:** 3.2 Highlight big ideas, patterns, and critical features using organizers and examples. 6.3 Facilitate organization with graphic organizers, prompts, and note-taking guides. 9.3 Develop self-assessment and self-reflection with tools, charts, and feedback displays.	Video: Collision theory (Oak National Academy)
Teacher's notes/ Reminders **ملاحظات المعلم والتذكير**	Hello Parents, Please encourage your child to review and practice Chemistry regularly each week using the resources available in their class Microsoft Teams - class material folder which is organized by week. The presentations, readings, videos, practice activities, etc. posted in Teams are the primary learning materials and should be used throughout the unit to support understanding and achievement. Effective preparation for formatives and summatives requires students to actively engage with the learning materials provided in their class teams. Thank you for partnering with us to support your child's learning and success.		

Week # الأسبوع#

Technology Integration دمج التكنولوجيا	TED-Ed, Khan Academy and Oak National Academy videos; scientific calculators; graph paper; d igital IB data booklet; slides shared after each lesson.
Cross-curricular links الروابط بين المواد الدراسية	**Mathematics:** gradients of tangents, graphs, units and significant figures. **Physics:** kinetic energy and the kelvin temperature scale. **Biology:** enzymes as biological catalysts.
External Assessment Integration دمج التقييمات الخارجية	IB DP Chemistry Papers 1 and 2: rate calculations from graphs, multiple-choice style hinge questions on collision theory, and structured questions using the command terms explain, predict, determine and deduce. Graph and error analysis skills also support the Internal Assessment (IA).
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	**Thinking:** interpreting graphs and justifying predictions. **Self-management:** rating themselves 1–5 against the success criteria at the start and end of lessons. **Social:** pair work; choose-your-challenge tasks offer stretch without pressure.
Standards المعايير	• **R2.2.1** The rate of reaction is expressed as the change in concentration of a particular reactant/product per unit time. • **R2.2.2** Species react as a result of collisions of sufficient energy and proper orientation. • **R2.2.3** Factors that influence the rate of a reaction include pressure, concentration, surface area, temperature and the presence of a catalyst.
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	**Guiding question:** How can the rate of a reaction be controlled? **Tool 1, 3, Inquiry 2:** Concentration changes in reactions are not usually measured directly. What methods are used to provide data to determine the rate of reactions? **Nature of science, Inquiry 3:** How can graphs provide evidence of systematic and random error? **Lesson 2:** What is the relationship between the kinetic molecular theory and collision theory?
Success Criteria معايير النجاح **(Measurable)**	● **Must (DOK 1):** define rate of reaction, and state that a reaction needs collisions of sufficient energy and proper orientation. ■ **Should (DOK 2):** calculate a rate from concentration–time data or a tangent, and explain the effect of each factor on rate using collision theory. ▲ **Could (DOK 3):** compare rates at different times or conditions, and justify predictions and sources of error from graphs. **Evidence:** hinge-question and exit-ticket responses, sorted into secure / nearly / not yet.
National Identity Domain مجالات الربط بالهوية الوطنية	Conservation الحفاظ علي البيئة Global Understanding التفاهم العالمي Heritage التراث المجال Area: Innovation and Infrastructure التطبيق Implementation: Controlling reaction rate (catalysts, temperature, concentration) makes industrial processes, such as fertiliser production at Ruwais, efficient and less wasteful (Conservation); science is built through global collaboration (Global Understanding); early Arab and Islamic scholars contributed to chemistry (Heritage). Students also link the high UAE summer temperatures to faster spoilage and reaction rates.